

Maintenance Schedule

Frac Pack

EPA/CARB

TF12V4000C1

TF12V4000C2

TF12V4000C3

Application Group 4D

MSE50794/02E



Power. Passion. Partnership.

System designation	Engine model	kW/cyl.	Application group
TF12V4000C1	12V4000T95R	140 kW/cyl.	4D, Frac operation (continuous operation with low load)
TF12V4000C2	12V4000T95	155 kW/cyl.	4D, Frac operation (continuous operation with low load)
TF12V4000C3	12V4000T95L	162 kW/cyl.	4D, Frac operation (continuous operation with low load)

Table 1: Applicability

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1 Maintenance Schedule

1.1 Preface

The maintenance schedule is applicable to products certified in accordance with U.S. emissions regulations and operated in the area of applicability of these regulations.

Information about the latest maintenance schedule is available at: <http://www.mtu-online.com>.

The following instructions must be observed for these products:

The maintenance schedule is an integral part of the emissions certification. Nonobservance of the maintenance instructions can result in violation of the statutory emission specifications. Modification or removal of mechanical or electronic components or the installation of additional components as well as the execution of calibration processes that might affect the emission characteristics of the product are prohibited by emission regulations. Emission-relevant components may only be serviced, replaced or repaired if components approved by MTU are used. Emission-relevant components are for instance control units, data records, sensors, injectors, exhaust flaps and all components of exhaust aftertreatment systems.

The maintenance system for MTU products is based on a preventive maintenance concept. Preventive maintenance facilitates advance operational planning and ensures a high level of equipment availability. The maintenance intervals and the scope of required maintenance activities are the result of operational experience and therefore represent recommendations. Severe operating and ambient conditions may require additional maintenance work and/or modification of the maintenance intervals.

Adherence to the specified maintenance intervals is essential to maintain product safety.

Maintenance activities are specified with intervals based on hours in operation and time limits. Whichever value is reached first shall apply.

The individual maintenance intervals are classified in accordance with the qualification levels QL1 to QL4.

QL1: Operational monitoring and maintenance work which does not require disassembly of the product.

QL2: Exchange of components and parts (corrective only, not part of the maintenance schedule).

QL3: Maintenance work which requires partial disassembly of the product.

QL4: Maintenance work which requires complete disassembly of the product.

Additional notes on maintenance and preservation:

For change intervals for fluids and lubricants as well as for preservation specifications, refer to the relevant instructions of the component manufacturers and the MTU Fluids and Lubricants Specifications. The latest version for drive systems is available online at: <http://www.mtu-online.com>, and for power generation systems at: <http://www.mtuonsiteenergy.com>.

Components which are not listed in this maintenance schedule must be serviced in accordance with Manufacturer's Instructions.

1.2 Allocation matrix application group 4D

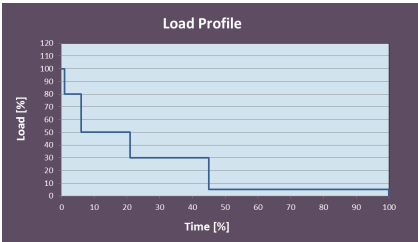
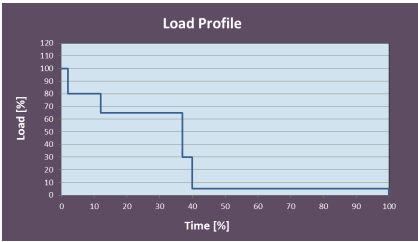
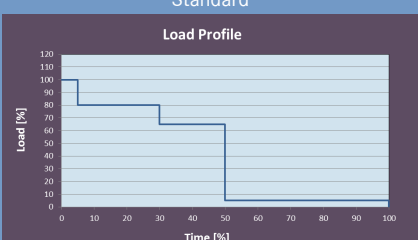
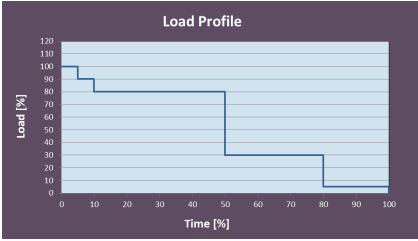
Load profile		12V/16V4000T95 / 12V4000T95x						
Load %	Time %	Load factor %	Load indicator %	121 to 140 kW/cyl.	141 to 163 kW/cyl.			
110 ₍₁₀₀₋₁₁₀₎	—	0 to 22	≤ 2	24,000	18,000	—	—	
100 ₍₉₀₋₁₀₀₎	1							
90 ₍₈₀₋₉₀₎	—							
80 ₍₆₅₋₈₀₎	5							
65 ₍₅₀₋₆₅₎	—							
50 ₍₃₀₋₅₀₎	15							
30 ₍₅₋₃₀₎	24							
Idle Speed ₍₀₋₅₎	55							
110 ₍₁₀₀₋₁₁₀₎	—	23 to 30	≤ 4	21,000	15,000	—	—	
100 ₍₉₀₋₁₀₀₎	2							
90 ₍₈₀₋₉₀₎	—							
80 ₍₆₅₋₈₀₎	10							
65 ₍₅₀₋₆₅₎	25							
50 ₍₃₀₋₅₀₎	—							
30 ₍₅₋₃₀₎	3							
Idle Speed ₍₀₋₅₎	60							
110 ₍₁₀₀₋₁₁₀₎	—	31 to 41	≤ 9	18,000	15,000	—	—	
100 ₍₉₀₋₁₀₀₎	5							
90 ₍₈₀₋₉₀₎	—							
80 ₍₆₅₋₈₀₎	25							
65 ₍₅₀₋₆₅₎	20							
50 ₍₃₀₋₅₀₎	—							
30 ₍₅₋₃₀₎	—							
Idle Speed ₍₀₋₅₎	50							
110 ₍₁₀₀₋₁₁₀₎	—	42 to 52	≤ 12	15,000	12,000	—	—	
100 ₍₉₀₋₁₀₀₎	5							
90 ₍₈₀₋₉₀₎	5							
80 ₍₆₅₋₈₀₎	40							
65 ₍₅₀₋₆₅₎	—							
50 ₍₃₀₋₅₀₎	—							
30 ₍₅₋₃₀₎	30							
Idle Speed ₍₀₋₅₎	20							

Table 2: Allocation matrix application group 4D

12V/16V4000T95 / 12V4000T95x							
Load profile		Load factor %	Load indicator %	121 to 140 kW/cyl.	141 to 163 kW/cyl.	-	-
Load %	Time %						
110 ₍₁₀₀₋₁₁₀₎	—	53 to 61	≤ 20	12,000	—	—	—
100 ₍₉₀₋₁₀₀₎	10						
90 ₍₈₀₋₉₀₎	15						
80 ₍₆₅₋₈₀₎	35						
65 ₍₅₀₋₆₅₎	—						
50 ₍₃₀₋₅₀₎	—						
30 ₍₅₋₃₀₎	30						
Idle Speed ₍₀₋₅₎	10						

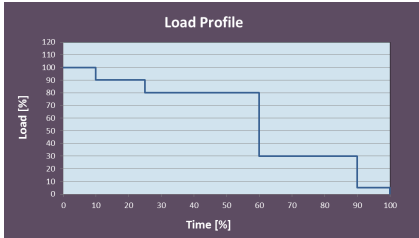


Table 3: Allocation matrix application group 4D

1.3 Maintenance schedule matrix 24,000 h

0 to 14,500 Operating hours

Task	Limit	Operating hours [h]																													
		Daily	500	1,000	1,500	2,000	2,500	3,000	3,500	4,000	4,500	5,000	5,500	6,000	6,500	7,000	7,500	8,000	8,500	9,000	9,500	10,000	10,500	11,000	11,500	12,000	12,500	13,000	13,500	14,000	14,500
Engine																															
W0800	-																														
W0505	1 w																														
W1024	1 w																														
W1008	2 a																														
W1009	2 a																														
W0500	-	X																													
W0501	-	X																													
W0503	-	X																													
W0506	-	X																													
W0507	-	X																													
W0525	-	X																													
W8646	-	X																													
W8644	-	X																													
W8645	-		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W1005	2 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W1001	2 a			X		X		X		X		X		X		X		X		X		X		X		X		X		X	
W1481	2 a			X		X		X		X		X		X		X		X		X		X		X		X		X		X	
W1675	2 a			X		X		X		X		X		X		X		X		X		X		X		X		X		X	
W1241	2 a			X		X		X		X		X		X		X		X		X		X		X		X		X		X	
W1575	1 a							X						X						X						X					
W1207	2 a			X				X						X						X						X					
W1551	2 a							X						X						X						X					
W1296	3 a							X						X						X						X					
W1817	18 a							X						X						X						X					
W1046	6 a									X								X								X					
W1525	6 a													X												X					
W1526	6 a													X												X					
W1610	18 a																		X												
W1577	18 a																		X												
W1689	18 a																		X												
W1006	18 a																		X												
W1636	18 a																		X												
W1011	18 a																		X												
W4184	18 a																										X				
W1818	18 a																		X												
W1250	6 a																										X				
W1251	6 a																										X				
W2012	18 a																										X				
W2001	18 a																										X				
W2006	18 a																										X				
W2007	18 a																										X				
W2009	18 a																										X				

w = weeks
m = months
a = years

Task	Limit	Operating hours [h]																													
		Daily	500	1,000	1,500	2,000	2,500	3,000	3,500	4,000	4,500	5,000	5,500	6,000	6,500	7,000	7,500	8,000	8,500	9,000	9,500	10,000	10,500	11,000	11,500	12,000	12,500	13,000	13,500	14,000	14,500
W2010	18 a																									X					
W2062	18 a																									X					
W2070	18 a																									X					
W2110	18 a																									X					
W1051	18 a																									X					
W1735	18 a																									X					
W1038	18 a																									X					
W1043	18 a																														
W1134	18 a																														
W1175	18 a																														
W3000	18 a																														
W3001	18 a																														
W3002	18 a																														
W3003	18 a																														
W3004	18 a																														
W3005	18 a																														
W3007	18 a																														
W3018	18 a																														
W3021	18 a																														
W3103	18 a																														
W3108	18 a																														
W3177	18 a																														
W3182	18 a																														
W1124	18 a																														
W1901	18 a																														
W1819	18 a																														
Transmission (ZF)																															
X4032	-	X																													
X4033	-	X																													
X4034	-	X																													
X4035	-		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
X4023	1 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
X4037	-			X		X		X		X		X		X		X		X		X		X		X		X		X			
X4042	-																														
W4146	-																														
Air compressor																															
W1784	-																X														
w = weeks m = months a = years																															

15,000 to 24,000 Operating hours

Task	Limit	Operating hours [h]																												
		15,000	15,500	16,000	16,500	17,000	17,500	18,000	18,500	19,000	19,500	20,000	20,500	21,000	21,500	22,000	22,500	23,000	23,500	24,000										
Engine																														
W0800	-																													
w = weeks																														
m = months																														
a = years																														

Task	Limit	Operating hours [h]																							
		15,000	15,500	16,000	16,500	17,000	17,500	18,000	18,500	19,000	19,500	20,000	20,500	21,000	21,500	22,000	22,500	23,000	23,500	24,000					
W0505	1 w																								
W1024	1 w																								
W1008	2 a																								
W1009	2 a																								
W0500	-																								
W0501	-																								
W0503	-																								
W0506	-																								
W0507	-																								
W0525	-																								
W8646	-																								
W8644	-																								
W8645	-	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X					
W1005	2 a	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X					
W1001	2 a	X		X		X		X		X		X		X		X		X		X					
W1481	2 a	X		X		X		X		X		X		X		X		X		X					
W1675	2 a	X		X		X		X		X		X		X		X		X		X					
W1241	2 a	X		X		X		X		X		X		X		X		X		X					
W1575	1 a	X						X						X						X					
W1207	2 a	X						X						X						X					
W1551	2 a	X						X						X						X					
W1296	3 a	X						X						X						X					
W1817	18 a	X						X						X						X					
W1046	6 a			X								X								X					
W1525	6 a							X												X					
W1526	6 a							X												X					
W1610	18 a			X																X					
W1577	18 a			X																X					
W1689	18 a			X																X					
W1006	18 a			X																X					
W1636	18 a			X																X					
W1011	18 a			X																X					
W4184	18 a																			X					
W1818	18 a			X																X					
W1250	6 a																			X					
W1251	6 a																			X					
W2012	18 a																			X					
W2001	18 a																			X					
W2006	18 a																			X					
W2007	18 a																			X					
W2009	18 a																			X					
W2010	18 a																			X					
W2062	18 a																			X					
W2070	18 a																			X					
W2110	18 a																			X					
W1051	18 a																			X					
W1735	18 a																			X					

w = weeks
m = months
a = years

Task	Limit	Operating hours [h]																							
		15,000	15,500	16,000	16,500	17,000	17,500	18,000	18,500	19,000	19,500	20,000	20,500	21,000	21,500	22,000	22,500	23,000	23,500	24,000					
W1038	18 a																			X					
W1043	18 a																			X					
W1134	18 a																			X					
W1175	18 a																			X					
W3000	18 a																			X					
W3001	18 a																			X					
W3002	18 a																			X					
W3003	18 a																			X					
W3004	18 a																			X					
W3005	18 a																			X					
W3007	18 a						X					X								X					
W3018	18 a																			X					
W3021	18 a																			X					
W3103	18 a																			X					
W3108	18 a																			X					
W3177	18 a																			X					
W3182	18 a																			X					
W1124	18 a																			X					
W1901	18 a																			X					
W1819	18 a																			X					
Transmission (ZF)																									
X4032	-																								
X4033	-																								
X4034	-																								
X4035	-	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X					
X4023	1 a	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X					
X4037	-	X		X		X		X		X		X		X		X		X		X					
X4042	-																			X					
W4146	-																			X					
Air compressor																									
W1784	-			X																X					
w = weeks m = months a = years																									